



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Friday, 11 September 2026, 3:37 PM**State** Finished**Completed on** Friday, 11 September 2026, 4:13 PM**Time taken** 35 mins 10 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main() {
5      int n;
6      if (scanf("%d", &n) != 1) return 0;
7
8      int *arr = (int *)malloc(n * sizeof(int));
9      int *dp = (int *)malloc(n * sizeof(int));
10
11     for (int i = 0; i < n; i++) {
12         if (scanf("%d", &arr[i]) != 1) return 0;
13         dp[i] = 1;
14     }
15
16     int max_len = 1;
17     for (int i = 1; i < n; i++) {
18         for (int j = 0; j < i; j++) {
19             if (arr[j] <= arr[i] && dp[j] + 1 > dp[i]) {
20                 dp[i] = dp[j] + 1;
21             }
22         }
23         if (dp[i] > max_len) {
24             max_len = dp[i];
25         }
26     }
27
28     printf("%d\n", max_len);
29
30     free(arr);
31     free(dp);
32     return 0;
33 }
34

```

	Input	Expected	Got	
✓	9 -1 3 4 5 2 2 2 2 3	6	6	✓
✓	7 1 2 2 4 5 7 6	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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